

Number Theory

Divisibility rules, remainders, prime factors.

Equations with integer solutions.

Divisibility / Remainders

1. How many four digit multiples of 9 consist only of four different odd digits
A 4 B 12 C 24 D 36 E 120

2. How many three digit numbers are divisible by 5 or 9?
A 200 B 220 C 240 D 260 E 280

3. The digit p, q, r, s, t are all different.
What is the smallest five digit integer ' $pqrst$ ' that is divisible by 1, 2, 3, 4 and 5?

4. Find the possible values of the digits p and q , given that the five digit number ' $p543q$ ' is a multiple of 45.

5. $30^{20} - 20^{20}$ is divisible by all the following values except for which one
A 10 B 25 C 40 D 60 E 64

6. If n is a multiple of 5 and $n = p^2q$, where p and q are prime numbers, which of the following must be a multiple of 25?
- A p^2 B q^2 C pq D p^2q^2 E p^3q
7. If both 11^2 and 3^3 are factors of the number $a \times 4^3 \times 6^2 \times 13^9$, what is the smallest value of a ?
- A 33 B 121 C 363 D 1089 E 3267
8. What is the highest positive integer power of 3, that divides $100!$
($100! = 100 \times 99 \times 98 \times \dots \times 3 \times 2 \times 1$)
- A 33 B 48 C 60 D 99 E 100
9. What is the remainder when $|5^{13} - 6^{14}|$ is divided by 10?
- A 1 B 2 C 5 D 6 E 9
10. How many different positive integers exist between 10^6 and 10^7 , the sum of whose digits is equal to 2?
- A 6 B 7 C 8 D 12 E 18

11. An integer n is divided by an integer k and leaves a remainder of 24.
When twice the integer n is divided by k the remainder is 11.
What is the value of k ?
A 13 B 35 C 37 D 59 E 18
12. What is the largest prime factor of $7^{20} + 7^{21} + 7^{22}$?
A 7 B 13 C 19 D 57 E 7^{20}
13. A natural number k is such that $k^2 < 2023 < (k + 1)^2$
What is the largest prime factor of k ?
A 5 B 7 C 11 D 13 E 19
14. The positive integer k is the smallest value such that $140k$ is a perfect square.
What is the sum of the digits of k ?
A 5 B 8 C 11 D 14 E 15
15. How many positive integers that are factors of $(2^3)(3)(5^2)$ are even?
A 6 B 12 C 16 D 18 E 24

Equations with integer solutions

1. How many ways can 75 be expressed as the sum of two or more consecutive positive integers?
A 1 B 2 C 3 D 4 E 5

2. Consider a sequence of five consecutive positive integers. How many such sequences are there, in which the sum of the squares of the first three integers is equal to the sum of the squares of the other two integers?
A 0 B 1 C 2 D 3 E 4

3. Given the equation $xy = 8x + 8y - 16$, where x and y are positive integers, find the smallest possible value of xy
A 0 B 16 C 64 D 224 E 240

4. How many ways are there to express $\frac{2}{15}$ in the form $\frac{1}{a} + \frac{1}{b}$ where a, b are positive integers with $a \leq b$
A 1 B 2 C 3 D 4 E 5

5. How many ways are there to express $\frac{5}{11}$ in the form $\frac{1}{x} + \frac{1}{y}$ where x, y are positive integers with $x \leq y$
A 0 B 1 C 2 D 3 E infinite

6. How many pairs of positive integers satisfy the equation $5a - ab = 9b^2$
A 0 B 1 C 2 D 3 E 4
7. How many pairs of integers satisfy the equation $(|a| - 3)(|b| + 4) = 12$
where a, b are integers
A 3 B 6 C 10 D 12 E 24
8. How many pairs of positive integers satisfy the equation $3x + 2y = 120$, where $x < y$
A 4 B 5 C 11 D 12 E 24
9. How many ways are there to express $\frac{1}{12}$ in the form $\frac{1}{m} + \frac{4}{n}$
where n is an odd integer less than 60
A 0 B 1 C 2 D 3 E 4
10. How many positive integer values of a are there such that $\frac{a + 220}{a + 4}$ is an integer
A 10 B 12 C 14 D 20 E 22

11. How many integer values of a are there such that $\frac{a}{1011-a}$ is an even integer
- A 1 B 4 C 8 D 10 E 12
12. How many integer solutions are there to the equation $|(|x| - 2023)| < 3$
- A 10 B 14 C 101 D 2020 E 2026
13. What is the greatest possible perimeter of a right-angled triangle with integer side lengths if one of the sides has length 12.
- A 15 B 20 C 60 D 84 E 120
14. If $3^x + 2^y = 985$ and $3^x - 2^y = 473$, what is the value of xy
- A 40 B 45 C 48 D 54 E 60

15. What is the largest integer n such that $\frac{50!}{(5!)^n}$ is an integer

A 5

B 12

C 15

D 20

E 25

16. What is the sum of all the possible integers n such that $n^2 + 6n + 24$ is a square number

A -12

B -6

C 0

D 4

E 20